II2202: Research Methodology and Scientific Writing

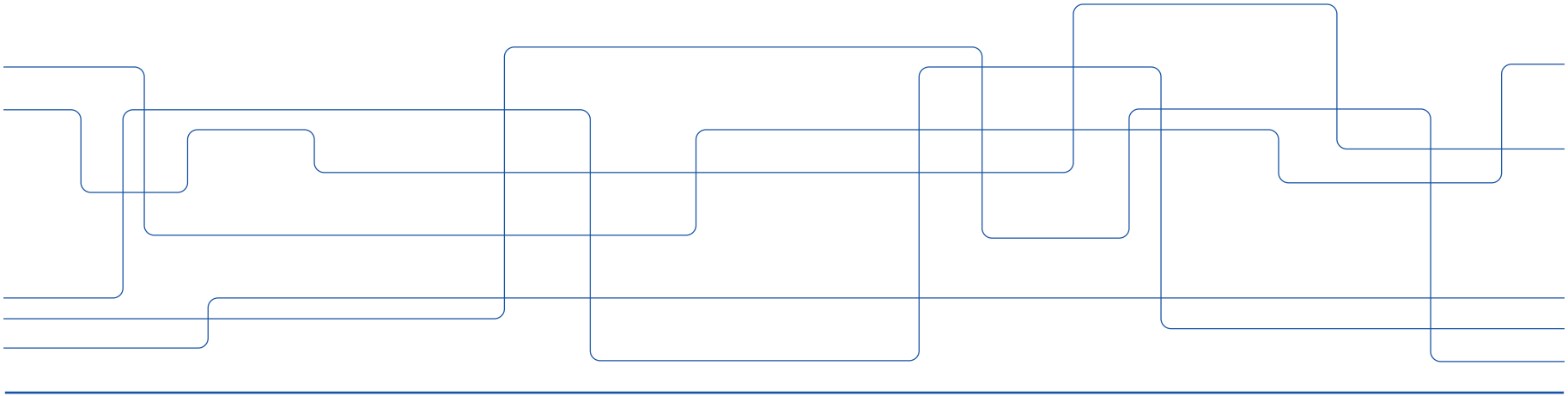
Introduction

Prof. Emil Björnson

kth.se/profile/emilbjo

School of Electrical Engineering and Computer Science (EECS)

Autumn 2026, Joint lecture for Period 1 and Periods 1-2 editions



Why take a course on "Research Methodology and Scientific Writing"?

To (1) gain a systematic overview and
(2) practice research methodology and scientific writing.

It will provide you:

- Enhanced project working skills during your education
- Improved degree project (thesis) quality
- Skills and understanding necessary for your professional life

Some of these required skills and understanding are described in Högskolelagen (1992:1434) - Higher Education Act

Our core ambition

The course should be fun!

- Learn what you need while doing a project on a topic you like!

Today's lecture overview

About this course

- Course schedule
- Teachers
- Language
- Prerequisites
- Goals
- Scope = topics covered
- Method
- Deliverables
- Assessment
- Equivalent courses / Examination / grading
- Literature
- Academic ethics and use of AI tools

Research Methodology

Using Canvas

Q&A

Course schedules

There are two versions of this course:

1. Period 1 (P1): August-October – for Software Engineering of Distributed Systems (TSEDM) and Communication Systems (TCOMM)
2. Periods 1 & 2 (P1P2): August-January – for ICT Innovation (TIVNM)

Note: There are different schedules and Canvas rooms for these course versions!

- Same content but different deadlines and supporting mentor teachers

See the course syllabus in Canvas – it gives you a **time-ordered** set of material, both content and assignments

You are advised to work ahead rather than chase deadlines!

Teachers associated with the course

Course responsible and Examiner

Prof. Emil Björnson kth.se/profile/emilbjo

Lectures and guest lecturers (via ~12h of videos)

Prof. em. Gerald Maguire, prof. Carl-Mikael Zetterling, prof. Magnus Boman, Assoc. prof. Markus Hidell, prof. Mark T. Smith, Dr. Jon-Erik Dahlin, Retired Professor Ellen M. McGee, Ph.D. (Long Island University), prof. em. Marilyn E. Noz (NYU), Niklas Olsson, …

Mentor Teachers

- P1 TSEDM: Anders Västberg
- P1 TCOMM: Ben Slimane, Vitaly Petrov
- P1P2 TIVNM (EIT) HCI: Jan Gulliksen
- P1P2 TIVNM (EIT) Autonomous systems: Emil Björnson, Ben Slimane
- P1P2 TIVNM (EIT) Data science: Henrik Boström
- P1P2 TIVNM (EIT) Cloud & Other: Irem Bor-Yaliniz, Nadia Bekkali, Ben Slimane

First meeting with your mentor teacher

Check in Canvas which section you belong to. It is the name of your mentor.

First mentor meetings:

- Anders Västberg: Aug. 28, 13-15
- Ben Slimane: Aug. 27, 10-12
- Vitaly Petrov: Aug. 28, 13-15
- Jan Gulliksen: Aug. 28, 10-12
- Emil Björnson: Sept. 2, 15-17
- Henrik Boström: Sept. 7, 13-15
- Irem Bor-Yaliniz: Sept. 4, 13-15
- Nadia Bekkali: Sept. 11, 10-12

No section assigned?
Send an email by 17:00 today to
Amna Irshad (P1),
Peng Zhang (P1P2) + cc to Emil

Who is the Examiner?

Highly Cited Researcher 2025

Clarivate

Emil Björnson

Professor

Works for
Communication Systems

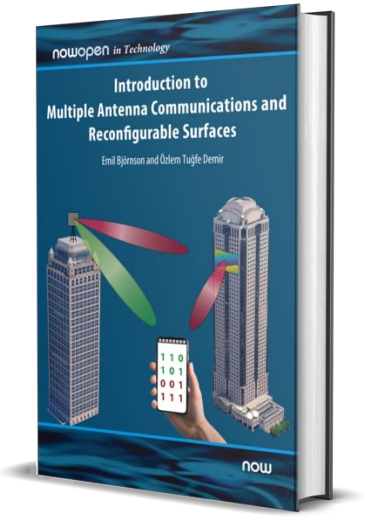
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Unit address
Malvinas Väg 10

E-mail
emilbjo@kth.se

Researcher ID
ORCID ID, Web of Science ResearcherID, Scopus ID, Google Scholar ID, ResearchGate





- Wireless communication researcher
  - Mathematical modeling and derivations
  - Computer simulations

- Published 4 textbooks, 200 journal papers, 225 conference papers, 30 filed patents
  - Host of the YouTube channel "*Wireless Future*" about wireless + research methodology

Language of the course

- The language used in the course is English
- There is plenty of pre-recorded lecture material
  - Most of the course videos have captions available in English, and these also have a transcript of the lecture material in English
  - Some of the material is also available in Swedish
    (there is a dual English and Swedish set of videos regarding sustainability)

Note: Some lectures refer to old document templates or assignment instructions. The main content is still meaningful but follow the new instructions when working!

Prerequisites

Eligibility (required)

Good knowledge of English and basic knowledge of information and communication technology

Goals, Scope, and Method

Goals

To help you be successful in your Master's courses, thesis project, and professional career

Scope

- research methodology
- scientific writing
- a scientific assignment

Method

Lectures (via videos), quizzes (in Canvas), advising by the teacher, literature (books, articles, websites, and tools),
*and a **staged project: from proposal to final report + opposition***

Specific Aims of this Course

- To develop the theoretical & practical skills to **plan, conduct, analyze, and present** (in oral and written form) **a scientific assignment** in the area of information and communication technology (ICT)
- To develop insight & understanding of **research methodology**, **ethics**, and **sustainability**

Learning Outcomes

After the course students will be able to:

- explain and apply scientific methodologies, methods, as well as techniques for scientific writing, and research methodology to prepare the writing of a scientific report, as well as a degree project.
- perform investigation and evaluation using methods, explain and take a position on [assess] the results, as well as list and summarize related work.
- apply the knowledge in scientific writing and research methodology and use this knowledge to write a scientific report and opposition report.
- identify and describe examples of sustainability related to the ICT-area
- verbally and in writing give examples of and explain societal and ethical aspects and sustainability in the ICT-area.
- perform an opposition

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*and a **staged project: from proposal to final report + opposition***

Contents

The course is divided into three major parts:

- research methodology
- scientific writing
- a scientific assignment

These parts are **integrated** into a project and examined by a project plan, a method description, a scientific report, and an opposition report.

You will give oral presentations and provide peer feedback at multiple stages of the project work.

Scope: Topics addressed in this course

Research methodologies

- Main focus on qualitative and quantitative methods
- Consider alternative methods & choose a suitable method

Scientific assignment

- Propose and motivate your investigation
- Apply quantitative/qualitative methods when collecting & analyzing your data
- Carry out your investigation

Scientific writing – reporting your results

- Improving your writing and oral presentation skills, including the use of tools enhance professionality
- Avoiding plagiarism
- Opposition – giving and receiving **feedback** (see [Stone and Heen 2014])

Research and professional ethics

Sustainability
(both generally and ICT specific)

Goals, Scope, and Method

Goals

To help you be successful in your Master's courses, thesis project, and professional career

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- research methodology
- scientific writing
- a scientific assignment

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Lectures (via videos), quizzes (in Canvas), advising by the teacher, literature (books, articles, websites, and tools),
*and a **staged project: from proposal to final report + opposition***

Lecture and Online Material

- First lecture – this lecture
  - After that, you will have regular meetings with your mentor teacher

- >12 hours of videos – with associated quizzes

  I suggest you watch these ***as soon as you can****, as* they will give you lots of material ***before*** you need it, rather than just waiting until the last moment.

  You can also watch them at 1.5x or at lower or higher speeds, as you prefer.

  There are also transcripts for most of the videos, so if you prefer to read, you can.

  The deadlines for finishing the quizzes are indicative, and the result does not affect your grade, but you must do all of them to pass the course. Some project-related assignments require you to have finished all the previous quizzes.

- A set of assignments and a document templates

Project-related Assignments

- Identify a research problem connected to literature ⇒ **Project plan**
- Research method description (theory, discussion about choice of method, selection of participants and data collection) ⇒ **Research plan**
- Scientific report ⇒ **Final Seminar** and **Final (written) Report**
- Opposition report ⇒ **(Written) Opposition before final seminar**

These will be developed in a staged fashion, and there will be mentor meetings, oral presentations, and peer reviews during this process.

Projects will be conducted in teams of **two** people (and never more than two).

Peer feedback and opposition will be performed **individually**.

Deliverables

Mistakes are an important part of learning!
Most tasks in the course allow you to train and make mistakes without “punishment”.

Project Proposal

Research Plan

Video Presentation of Research Plan, Ethics, and Sustainability

Peer Feedback on Video Presentation

Qualitative Exercise with Peer Feedback

Quantitative Exercise with Peer Feedback

First draft of Final Report and Presentation

Provide the Second Draft of your Final Report to Your Opponents

→ Final grade (indirectly)

Written Opposition Report

Final Seminar with Oral Opposition

Final Written Report

Self-Reflection on the Project Work and Learning

→ Final grade (directly)

Project Groups

- Form two-person groups with one other person from the same “section”.
  - Section = Students preassigned to a specific mentor teacher
  - Do so by August 28 (P1 version) or September 4 (P1P2 version)
  - Ideally, before your first mentor meeting.
  - Register your group in Canvas (check the instructions in the Syllabus)

- Students who do not select and register a teammate will be assigned to a group by the responsible teacher the week after the deadline.

No section assigned? Send an email by 17:00 today
Amna Irshad (P1), Peng Zhang (P1P2) + cc to Emil

Choosing a Project Topic

- You choose your own research topic.
  - Make sure the scope is limited enough for this course.
  - Build on things that you are good at and/or interested in!
- Discuss with your mentor teacher whether the topic is appropriate.
  - Some iterations and polishing are usually needed
  - One might even need to throw out an idea and replace with a different one

- For inspiration, topics from previous years can be found under the modules "Examples of titles from previous P1 reports" and "Examples of titles from previous P1P2 reports"

Performing Peer Review/Feedback

- Mastering the art of providing constructive feedback is an essential skill emphasized in this course. You will perform peer feedback throughout this course.
- This part is an INDIVIDUAL ASSIGNMENT.
- Non-graded feedback: On a video presentation, qualitative exercise, and quantitative exercise
- Graded feedback: Written and oral opposition on a final project report

General points:

- You should be able to access work for peer review latest 24 h AFTER the submission deadline, and you then have two working days to complete the review (unless otherwise stated).
- Your peer feedback should be constructive to help the group that submitted the assignment to improve their work before the next submission round.
- Providing only cheerful support does not fulfill this purpose.
- Grades of other groups can never be affected by critical reviews; the grading will be done exclusively by the teacher, so there is no reason to be “kind” in your reviews (but you should always keep the tone polite and constructive).

About Project Report Writing

- In today's world of information overload, an appropriate format is a short report that highlights the key findings and messages.
  - Details or less important findings can be left out or placed in an appendix.
- High-impact scientific journals (Nature, Science) typically restrict word count to **< 3000 words** (~5 pages text; this excludes methodology and references) and **a maximum of 6 figures or images**.
  - This is too condensed for our purposes, as the material will be used for grading.
- In this course, the final report has a strict page limit of 10 pages, excluding the title page and references. We consider conciseness to be a sign of quality.

Example: Summarize work in an abstract

Perform in an iterative update loop:

1. Attempt a five-sentence **abstract** of the scientific publication you are planning:
    1. Introducing the bigger perspective.
    2. Describe the specific challenge(s) or need(s) you aim to address.
        "The state of the art is limited by …"
    3. State the specific research questions (w or w/o indication of method).
        "Here we study …"
    4. State the key findings, quantitatively if possible. (May need more than one sentence.)
        "Our key findings are …"
    5. Outlook/impact statement: how can the findings benefit the world / why should your reader care about your results?
        "The significance of these findings lays in …"

Not being able to formulate concisely can be a sign of not fully understanding the key importance. ***Focus!*** *Cut away what is not important.*

Similar approach in other sections:
Tell the reader what section will contain.
Tell the important things.
Summarize the main points.

On instructions and timeliness…

- If you're uncertain or have specific queries about any course-related aspect (e.g., unclear instructions, deadlines, expectations…), contact your mentor teacher.
  - Your understanding is paramount, and we're here to support you.
- It is essential that you are proactive.
  - The teachers will not tell you what to do, but you need to take the initiative and ask for advice!

- In cases where guidelines aren’t followed, or deadlines aren’t met, the respective assignments may be marked as “incomplete” or "fail”, which might affect your grade negatively, as making plans and following your own plans are part of the learning goals.
- **Specifically:** Groups that submit late will typically not receive peer review. (They remain obliged to provide peer review on other work.)

Assessments

Formative assessments

- Your mentor teacher gives such feedback in meetings and on your reports.
- You will also have some formative assessments based on your responses to the quizzes for the videos.
- Peer-reviewing other students' drafts, presentations, and exercises along the way.

Summative assessments

- Written opposition before the final seminar,
  Final seminar (oral presentation and oral opposition), and Final report
- The work process (ability to utilize feedback to make enhancements) and the final report.

Grade Levels

PRO1 - Project Assignments, 7.5 credits

Grade scale: A, B, C, D, E, Fx, F

Note that Fx is **not** a final grade.

Note that the final grades are assigned **individually**; therefore, if your teammate is not contributing equally:
(1) discuss this with them,
(2) discuss this with the mentor teacher working with your section,
(3) disclose this in your self-evaluation at the end of the course.

Grading Criteria

1. Research problem, questions, theory, and related work
2. Research methodology, design, and execution
3. Results, analysis, discussion, validity, and conclusions
4. Scientific writing and written presentation
5. Research process and use of feedback
6. Ethics, societal aspects, and sustainability
7. Final seminar and individual oral examination
8. Written and oral opposition

Quality of final report (Group level)

Final examination (Individual level)

How is the Grading Done?

There is a detailed “Assessment and Grading Rubric” in Canvas

- It describes the expected performance to reach Excellent (A), Good (C), and Sufficient (E) for each criterion. B and D lie between the descriptions.

Your mentor grades your performance on each of the criteria

- If all grades are at a passing level (E or higher), you will pass the course, and the final grade is selected by the examiner.
- To reach the final grade X, most of the eight criteria should be on level X or better, and no grades should be *significantly* worse.

Hint: It is common for both group members to receive the same grade, but not necessary, so team up with someone with a similar level of ambition.

Further Grading Information

If you fail to meet the learning objectives or your project has serious errors, the grade will be an "F".

If you are close to passing, but not at the passing level **(grade Fx)**, then you will be offered the opportunity for “komplettering”, i.e., submitting a revised version of the deficient components of the project assignments, which will be evaluated. You will receive an “E” if the revised components are deemed sufficient.

Note that there is **no** opportunity to raise your grade once you have a grade of “E” or higher.

Literature

Course website (Canvas)

- Video lectures
- Literature: list of books and a paper
- Useful tips, tools, and other information (see “Modules”)

Project specific material

- You need to develop your skills in **finding**, **evaluating**, and **summarizing related work**
- This includes finding the **background material & tools** that you need (1) to understand and use in order to carry out your project and (2) to support your data collection, analysis, and presentations

Learn and Follow Proper Academic Ethics

You will search and read scientific literature in conjunction with your projects. Make sure that you properly reference your sources in your report - keep in mind the relevant **KTH policies.**

In particular:

- If you use someone else's words, they must be clearly indicated as a **quotation** (***with a proper citation***).
  - We use a digital system to identify potential plagiarism
- Individual figures have their own copyrights, so if you are going to use a figure/picture/… from a source, you need to **both cite this source & have the copyright owner's permission to use it**.
  - It is okay to redraw a figure in a way that resembles the original

Ethics, Rights, and Responsibilities

At KTH there is a policy of zero tolerance for **cheating, plagiarism, etc.** **F**or details, see relevant KTH policies, such as https://www.kth.se/en/student/studier/rattigheter-och-skyldigheter/rattigheter-och-skyldigheter-1.1148520

Before starting to write, read the page about **cheating and plagiarism** at https://www.kth.se/en/student/stod/studier/fusk-1.997287

We have designed the course and its instructions to help you follow these professional practices.

See also the book: Jude Carroll and Carl-Mikael Zetterling, *Guiding students away from plagiarism* [Carroll 2009]

Guideline for Using Generative AI Tools

Generative AI has become a useful tool for researchers, but it must be controlled by humans.

The following pragmatic guidelines apply to this course:

1) You **may** use a generative AI tool to quickly gain an overview of a concept or topic; see the next slide for examples of allowable prompts.

2) You **may** use a generative AI tool to help you improve the language or argumentation in *your text*, but you should treat it as a colleague who gives you constructive feedback. The feedback is not always correct and can be quite generic, as the AI is risk-minimizing. Use its feedback to make text/presentation changes you agree on, but you are not allowed to include auto-generated text in your submissions – that is a form *of plagiarism*.

3) You **may** use a generative AI tool to help you program and process data, but you must always verify the outcome and be able to explain all the details in the final examination.

4) You **must** explain how you used generative AI in the dedicated sections of your reports.

Note that generative AI tools frequently produce incorrect details and add unnecessary/irrelevant information. As always, you are responsible for the content of **your** work, **and it must represent your own work**!

1. Using Generative AI for Information Finding:
Gain a rapid overview of a concept or topic

Examples of allowable prompts for gaining a rapid overview:

1. "What are the key principles and applications of artificial intelligence in healthcare?"

2. "Provide an overview of the major programming languages used in web development and their features."

3. "Explain the fundamental concepts and techniques in data mining and their role in business analytics."

In these prompts, you are seeking general information or an overview of a specific subject. They are *not* asking the AI to generate specific answers or complete their assignments.

Keep a record of all prompts that provided useful information. If you want to use any AI-provided information in your report, you need to find original sources to support it.

2. Using Generative AI to Improve Language

You must use the following prompt:

"Background: *I am a student in the course "II2202 Research Methodology and Scientific Writing" at KTH, and I am using this tool for a project. My teacher allows me to use AI tools to improve the language and articulation in my writing, but not to generate new ideas.*

Instruction: *Please suggest how to enhance the "input text" below, pointing out any errors, and provide feedback on the clarity, spelling, sentence structure, and grammar. List all spelling and grammar errors. If appropriate, list up to three suggestions for a change in style, coherence, or logical flow. Provide a suggestion for the rephrased text.*

Specific: *Focus on refining my existing ideas. Ensure no ideas are removed or lost. Ensure that the intellectual content is unchanged and no new concepts are introduced.*

Input text: *[insert paragraph text here]"*

3. Using Generative AI to Create Illustrations

You are allowed to use generative AI to create illustrations for your presentation and report, but you must then write a detailed prompt that controls the content, and you are responsible for the correctness of the end result.

Keep in mind:

- Generative AI tends to produce overly detailed infographics, which are not suitable for a scientific report or presentation. Refine the result until it matches the style that you have seen in scientific papers.

Note: If you include an AI-generated illustration in your report, then you must include the prompt in the appendix.

4. Reporting of use of AI Tools

In many of the assignments, you are requested to declare if and how you used AI tools. Read the instructions carefully, as the permitted uses vary across them.

Examples:

- In the research plan, you should describe how you *intend* to use AI in your project.
- In your final report, you should explain how you *actually used* AI in your project, and provide details in an appendix.

Note: Grammarly and similar tools that correct typos/grammar inline can be used without listing corrections.

5. No Autogenerated Text

Examples of prompts that are NOT allowable:

1. "Write a summary of the advantages and disadvantages of using AI in autonomous vehicles."

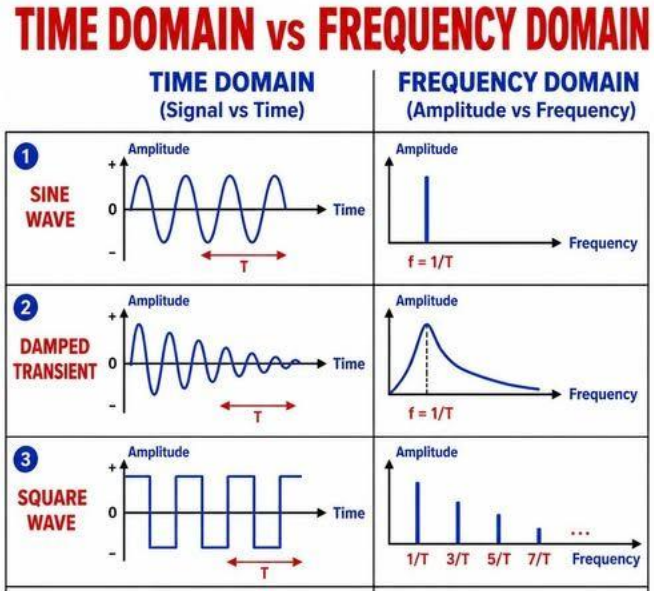
2. "Compose an essay on the ethical implications of AI in social media algorithms."

3. "Provide a detailed analysis of the impact of AI on climate change mitigation."

These prompts are requesting the AI to generate specific content or complete your assignment. They go beyond the scope of using generative AI as a tool for gaining a rapid overview or giving feedback on your writing, which is not allowed according to the instructions given for this course.

Warning about use of Generative AI

- Frequently, the prompts are stored by the AI tool provider – be careful if intellectual property protection is important!
- Frequently, the information provided is incorrect or insignificant
- An example of “AI slop”:



Further warning about use of Generative AI

- In my experience, proper use of AI tools can help students elevate their projects from D to C, but deep human effort is required to reach A and B in this course.
- The less experienced you are with a topic, the harder it gets to make good use of the output from Generative AI. We have seen a tendency for below-average students to receive lower scores than usual due to overreliance on these tools.
- At least 33% of the language suggestions are incorrect/misleading, so one must actively assess which feedback to consider and what to reject.

The Qualitative Exercise: AI tools

In the assignment called “Qualitative exercise”, you will practice writing interview questions and then test how well the questions work by interviewing the AI.

For this specific exercise, you only need to follow the instructions. *No further AI prompts need to be included.*

Some of the ways to fail the course

- **Choosing an inappropriate topic (for example, one that cannot be carried out during the time period of the course) or not collecting sufficient data to achieve your desired confidence level**
- Failing to present appropriate arguments in your report; not discussing relevant related work; improper or incorrect analysis of your data; not considering the social, ethical, and sustainability aspects of your topic; failing to consider if your results are valid or not. The teacher will point such things out when reading your report, and failure to make the required changes is associated with failing.
- Misconduct – including plagiarism
- Failure to submit material on time
- Failure to give useful feedback to others; failure to utilize the feedback that you get
- …

Course Development in 2026

We have made significant enhancements compared to last year's course round

- The templates for all reports have been revised for better clarity and mutual alignment
- The grading rubrics are now specified more clearly
- The policy for the use of Generative AI has been adjusted

Please report to the examiner if any ambiguities arise from this!

Contacting Us

Note the name of your mentor teacher:

- Anders, Ben, Vitaly, Jan, Emil, Henrik, Irem, or Nadia

When to contact your mentor teacher: be proactive!

- Schedule and attend meetings with your mentor. Prepare things to ask!
- The research topics are selected by the students. Discuss your topic with your mentor!
- Contact your mentor if you do not understand aspects of the course content.
- Talk to your mentor teacher to confirm the date for the final presentation!

When to contact the TAs (Amna Irshad in P1, Peng Zhang in P1P2) with cc to the examiner Emil:

- For course administrative issues
- Problems with Canvas
- Second line support for mentors.

Today's lecture overview

About this course

Research Methodology

- What is research?
- Why do we do research?
- How do we do research?
- What shall we research?
- How can we be sure of what we know?
- Where did we gain our knowledge?
- Can we believe our knowledge?

Using Canvas

Q&A

Research: What/Why/How?

What is research?

Gathering of data, information, and facts for the advancement of knowledge

Why do we do research?

Solve problems, prove new ideas, develop new theories, to achieve (attain) knowledge, to establish new facts, establish truth, refute falsehood, driven by curiosity

How do we do research?

Systematically collect and analyze data to increase our understanding of the phenomenon we are studying

Conduct the research ethically

What shall we research?

Frequently, the research is part of **an assignment** for a course, degree program, or your job.

Research should have a **worthwhile purpose**, hence it **should** have some **benefit** to society, research subjects, and/or gain useful knowledge

There are also ethical issues regarding whether to conduct specific research and how to disseminate the outcomes to others.

Högskolelagen (1992:1434)
Higher Education Act [*Högskolelag*]

"The Swedish Government has furthermore added an important goal for higher education and research that may influence choices of scientific perspectives and problems:
It shall **support a sustainable development that creates a good, healthy environment for this and future generations, economical and social welfare and justice** (Högskolelagen - Higher Education Act)."

CODEX, Rules & guidelines for Research,
The humanities and social sciences

Formerly at http://www.codex.vr.se/en/forskninghumsam.shtml

[Emphasis added to the quote above by GQMJr]

The Researcher's responsibilities

You should:

- **Contribute to the advancement of knowledge** (for example, the understanding of the phenomenon under study) and
- **Communicate that new knowledge/understanding to others**

These are in addition to the researcher's other economic, social, ethical, …, responsibilities.

Research Methods and Methodologies

1. **Select the most suitable** methods and methodologies *to answer your research questions*
2. **Correctly apply** the/these method/s and methodologies

Avoid picking methods and methodologies that do **not** match and/or complement each other

Keep in mind that different research questions require different research methods, even if the overall research aim is the same

Methodology

"1: a body of methods, rules, and postulates employed by a discipline: a particular procedure or set of procedures

2: the analysis of the principles or procedures of inquiry in a particular field"

http://www.merriam-webster.com/dictionary/methodology

Research Methodologies

Methodologies provide:

- Overarching theoretical and philosophical frameworks that guide your research (theoretical perspective, principles)
- Guidelines for solving your research problem, i.e., what are the correct procedures to find reliable solution(s)

Choice of methodologies concerns how we use logic, establish reality, assign values, and what we consider knowledge

Choice of research methodology/methodologies is important at the **start** of the research, as they must be aligned with the problem at hand

Research Methods

Methods are the **tools**, **techniques**, or **processes** used:

- Offer specific techniques to collect your data:
  survey, interviews, case studies, observations, experiments, etc.
- Define the steps needed to conduct your research and to find your solution
- Support your carrying out high-quality, valid, and trustworthy research – reproducibility is key

Your chosen research method(s) bring us to the **end point** of our research.

Why Learn Different Research Methods?

So that you can choose the **appropriate** method(s) for *your problem*.

The choice of method is going to give you a direction and systematic way of conducting your research.

It will also determine how you collect your data, analyze it, and draw conclusions.

We will focus on two major classes of methods:
Qualitative and **Quantitative** methods

Quantitative and Qualitative research methods (in ICT)

Quantitative

- Measurements, numbers, data, statistics, … ⇒ models, theories, verification/refutation of hypotheses
- Logical, objective, and quantifiable procedures

Qualitative

- Observations, interviews, … ⇒ How does the user feel, Which choices will they make, Explanations for actions, …
- Subjective, interpretative, and seeking meaning

Triangulation = combining both methods in a complementary fashion

Examples of Quantitative and Qualitative research methods

At different levels:

- **Research methods**: experimental, conceptual, descriptive, empirical, …
- **Research approaches**: deductive (top-down), abductive (bottom-up), inductive (fill holes)
- **Research strategies & designs**: experimental, case study, survey, exploratory, *ex post facto*, action research, grounded theory, …
- **Data collection methods**: Experiments, questionnaires, observations, interviews, language & text, …
- **Data analysis methods**: statistics, computation, coding, …
- **Quality assurance**: replicability, validity, reliability, …
- **Presentation**: oral/written/video/… presentation of results and conclusions

See [Håkansson 2013]

Bounded choice of research methods (for this course)

- Research in the form of a literature review is not allowed, as students seem to confuse a literature review with simply a literature study. However, you are expected to describe relevant prior literature in Section 1 of your reports.
- Similarly, research projects are not allowed to be surveys, as students generally do not understand just how difficult it is to do a survey properly (practically, ethically, and statistically). If one is not already deeply knowledgeable in the field, a survey typically becomes nothing more than a simple literature study with no new intellectual content.
- *Ex post facto* research is allowed if there is a large enough dataset and the students have a new way to examine the data.

Trust but Verify!

How can we be sure of what we know?

Where did we gain our knowledge?

Can we believe our knowledge?

You have to:

- Read critically and widely
- Examine the reliability and validity (integrity, authenticity, credibility, and criticality) of the results that are presented
- Actively question results – which may include repeating the experiments yourself
- Examine the ethics of what you and others do
- Be self-critical, so that you do not mislead yourself or others!
- **Consider asking AI tools to criticize your work** (but be critical of their criticism)

Today's lecture overview

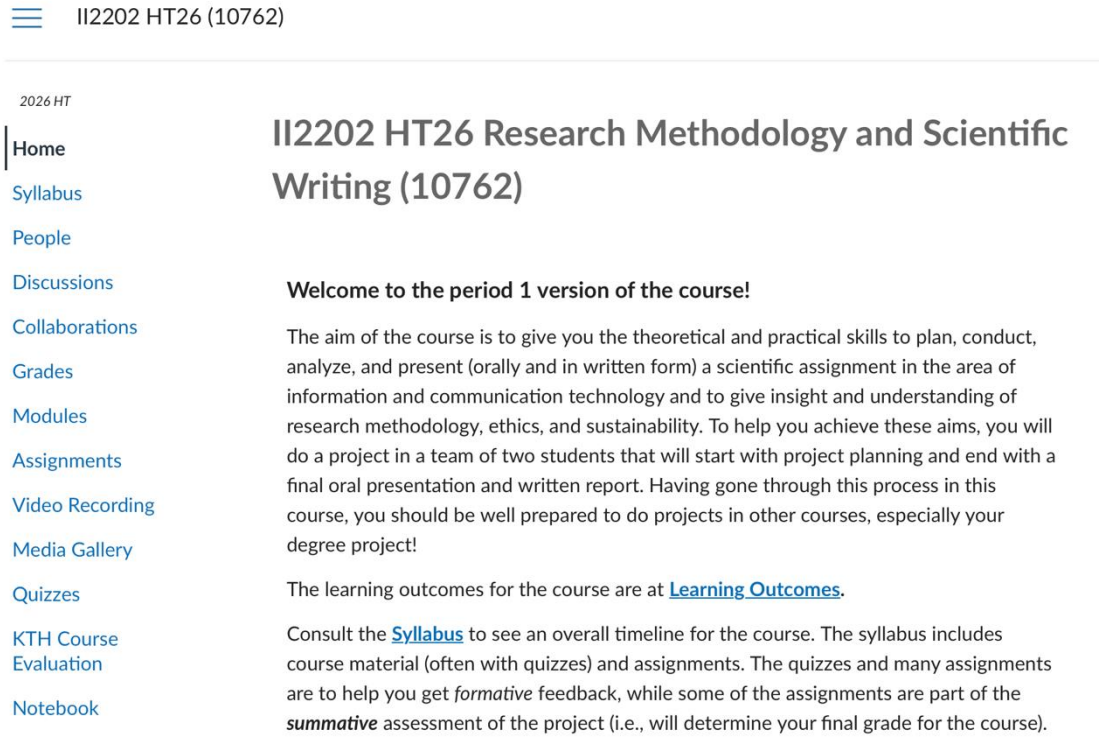
About this course

Research Methodology

Using Canvas

Q&A

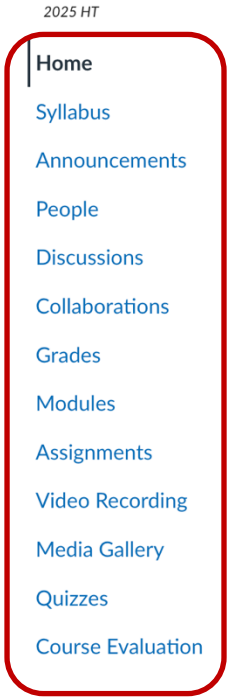
Navigating the Canvas course page



P1 https://canvas.kth.se/courses/63856

P1P2 https://canvas.kth.se/courses/63857

Navigating the Canvas course page



Webpages for the course

This Canvas course room is where the entire course is administered for registered students.

- You can find examples of previous project reports for groups that received a grade A here.
- The official course page is www.kth.se/student/kurser/kurs/II2202?l=en
- The course memo page is https://www.kth.se/kurs-pm/II2202/II220220252-50453-50517

Navigation within the Canvas course room

The easiest way to navigate the course room is through the **Syllabus**, which shows the time sequence of what to do.

The Syllabus links to all material in the sections Modules, containing all course material including videos and transcripts, and Assignments, containing all assignment descriptions. There is no immediate need to browse through the very large body of material in the sections "Modules" and "Assignments".

Discussions is a plac

People shows where

Grades shows your p

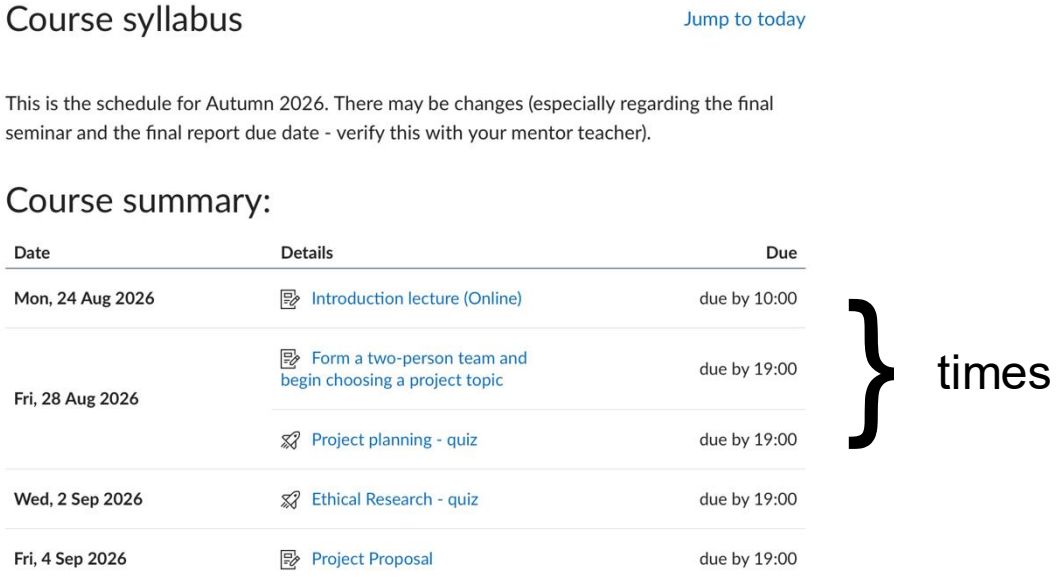
Video Recording has

Announcement will be sent out during the course – make sure to have email notifications turned on!

Syllabus page (P1 and P1P2) - Course timeline

This is the place to look for a time-ordered set of activities for you during the course.

The times for the course material are meant to suggest an **ordering** of the material, and as with the deliverables, represent a deadline by which you should have **completed** the assignment.



Course syllabus

Jump to today

This is the schedule for Autumn 2026. There may be changes (especially regarding the final seminar and the final report due date - verify this with your mentor teacher).

Course summary:

Date	Details	Due
Mon, 24 Aug 2026	Introduction lecture (Online)	due by 10:00
Fri, 28 Aug 2026	Form a two-person team and begin choosing a project topic	due by 19:00
	Project planning - quiz	due by 19:00
Wed, 2 Sep 2026	Ethical Research - quiz	due by 19:00
Fri, 4 Sep 2026	Project Proposal	due by 19:00

} times

Language of the Canvas page

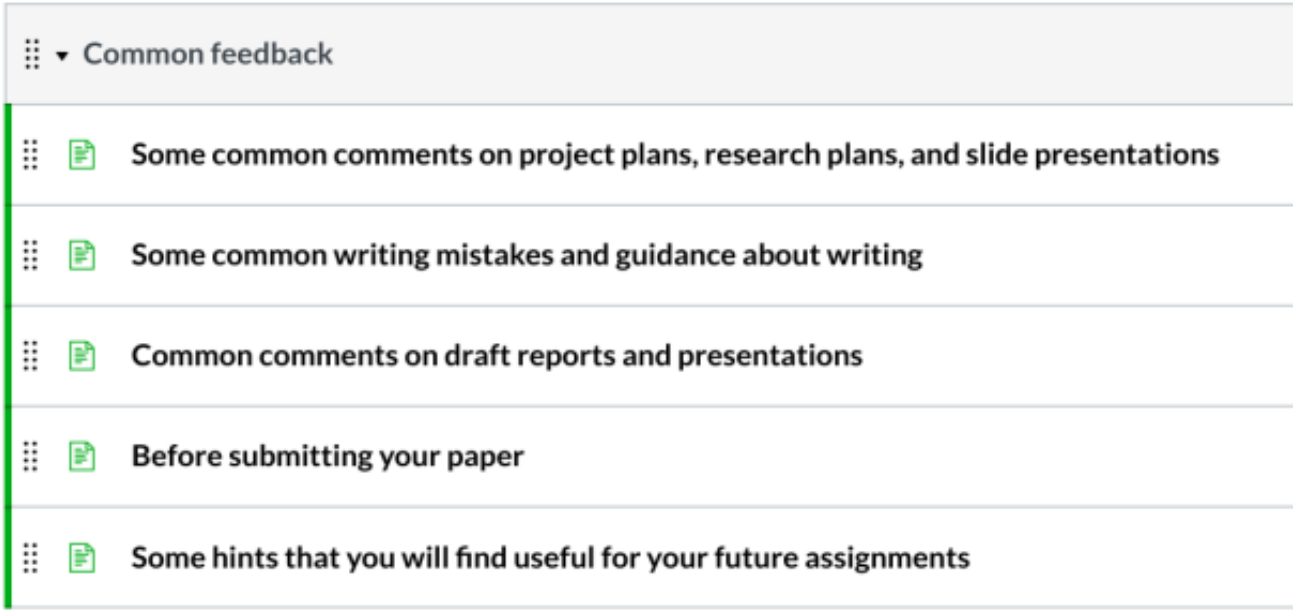
You can change KTH's default Canvas language from Swedish to other languages.

- Konto ≡ Account
- Inställningar ≡ Settings
- Redigera inställningar ≡ Edit settings
- Språk ≡ Language ---- choose the language you want
- Uppdatera inställningar ≡ Update settings

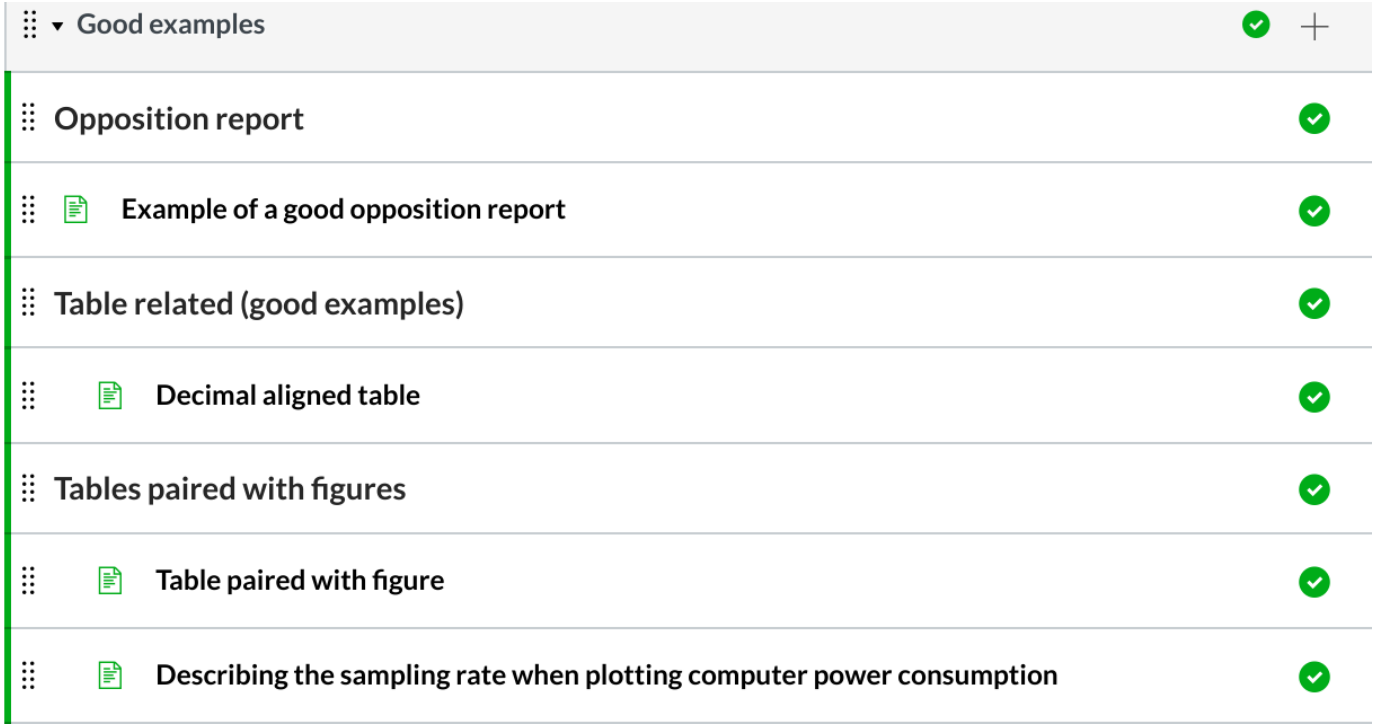
More info about Canvas: https://guides.instructure.com/

FAQs and Common feedback (under “Modules”)

There is a “Frequently ask questions (FAQ)” page for the course and a module with some common feedback:

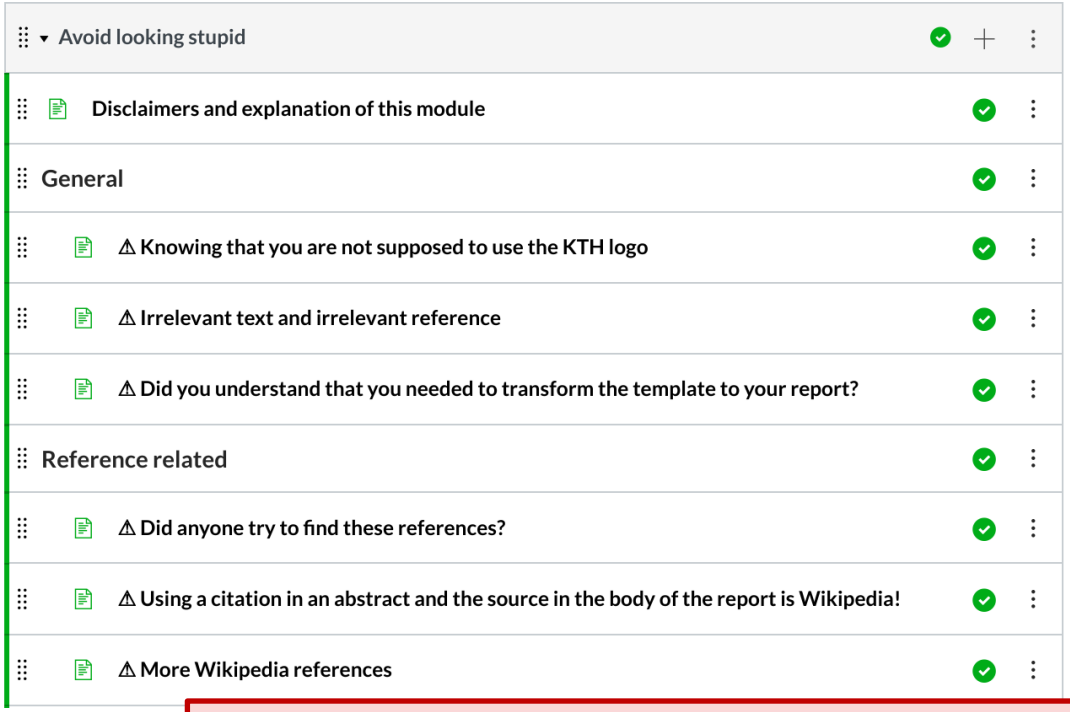


Good examples (under “Modules”)



...

Avoiding looking stupid (under “Modules”)



…

Is a link broken? Contact the course responsible (Emil)

Key References

[Carroll 2009] Jude Carroll and Carl-Mikael Zetterling, *Guiding students away from plagiarism/Hjälp studenterna att undvika plagiering*. Stockholm, Sweden: KTH Learning Lab, 2009, ISBN: 978-91-7415-403-0 (print) [Online]. Available: http://urn.kb.se/resolve?urn=urn%3Anbn%3Ase%3Akth%3Adiva-83704

[*Högskolelag*] Sveriges Riksdag, *Högskolelag*, vol. SFS 1992:1434. 1992 [Online]. Available: https://www.riksdagen.se/sv/dokument-lagar/dokument/svensk-forfattningssamling/hogskolelag-19921434_sfs-1992-1434

[Håkansson 2013] Anne Håkansson, 'Portal of Research Methods and Methodologies for Research Projects and Degree Projects', in *Proceedings of the International Conference on Frontiers in Education : Computer Science and Computer Engineering FECS'13*, Las Vegas, Nevada, USA, 2013, pp. 67–73 [Online]. Available: http://urn.kb.se/resolve?urn=urn:nbn:se:kth:diva-136960

[Stone and Heen 2014] Douglas Stone and Sheila Heen, *Thanks for the feedback: the science and art of receiving feedback well (even when it is off base, unfair, poorly delivered, and frankly, you're not in the mood)*. New York, New York: Viking, 2014, ISBN-13: 978-0670014668. http://stoneandheen.com/

[Sällberg 2015] Kristian Sällberg, 'A Data Model Driven Approach to Managing Network Functions Virtualization: Aiding Network Operators in Provisioning and Configuring Network Functions', Master's thesis, KTH, School of Information and Communication Technology (ICT), Stockholm, Sweden, 2015, TRITA-ICT-EX-2015:159. [Online]. Available: http://urn.kb.se/resolve?urn=urn:nbn:se:kth:diva-171233

Zotero group for the course: https://www.zotero.org/groups/kth-ii2202

Today's lecture overview

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